# Chapter 9, Problem 79P

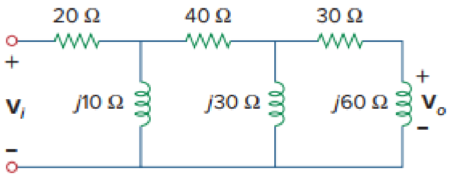
## Problem

(a) Calculate the phase shift of the circuit in Fig. 9.82.

(b) State whether the phase shift is leading or lagging (output with respect to input).

(c) Determine the magnitude of the output when the input is 120 V.

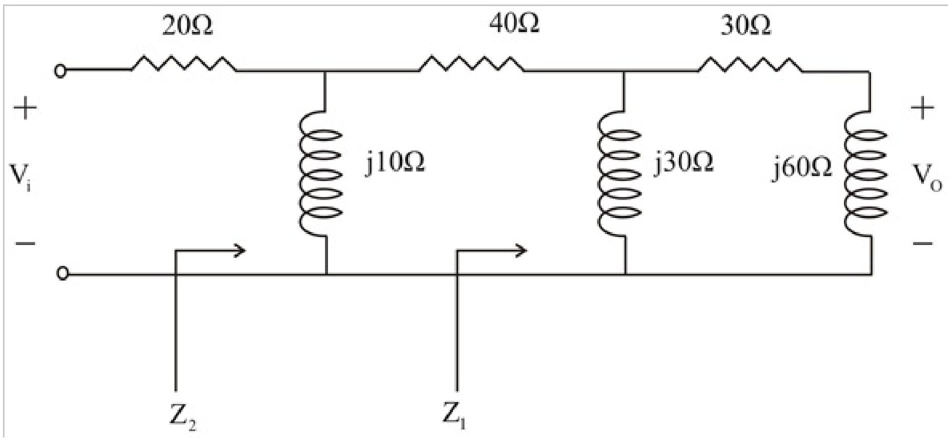
**Figure 9.82:**




## Step-by-step solution

**Step 1** of 5

(A) Consider the circuit shown.




**Step 2** of 5

$$Z_1 = j30 \parallel (30 + j60) = \frac{(j30)(30+j60)}{30+j90} = 3 + j21$$

$$Z_2 = j10 \parallel (40 + Z_1) = \frac{(j10)(43+j21)}{43+j31}$$

$$= 1.535 + j8.896 = 9.028\angle 80.21°$$

Let $V_1 = 1\angle 0°$

$$V_2 = \frac{Z_2}{Z_2+20}V_1 = \frac{(9.028\angle 80.21°)(1\angle 0°)}{21.535+j8.896}$$

$$V_2 = 0.3875\angle 57.77°$$

$$V_1 = \frac{Z_1}{Z_1+40}V_2 = \frac{3+j21}{43+j21}V_2$$

**Step 3** of 5

$$= \frac{(21.213\angle 81.87°)(0.3875\angle 57.77°)}{47.85\angle 26.03°}$$

$$V_1 = 0.1718\angle 113.61°$$

$$V_0 = \frac{j60}{30+j60}V_1 = \frac{j2}{1+j2}V_1 = \frac{2}{5}(2+j)V_1$$

$$V_0 = (0.8944\angle 26.50°)(0.1718\angle 113.6°)$$

$$V_0 = 0.1536\angle 140.2°$$

Therefore the phase shift is $140.2°$.

**Step 4** of 5

(B) The phase shift is leading.

**Step 5** of 5

(C) If V1 = 120 V, Then,

$$V_0 = (120)(0.1536\angle 140.2°)$$

$$= 18.43\angle 140.2°\ V$$

And the magnitude is 18.43 V.